Tribhuvan University

Faculties of Humanities and Social Sciences

FARM2DOOR

A PROJECT REPORT

Submitted to

Department of Computer Application

Orchid International College

In partial fulfilment of the requirements for the Bachelors in Computer Application

Submitted by

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Orchid International College

SUPERVISOR RECOMMENDATION

I hereby recommend that this project report prepared under my supervision by Pasang Tasi Sherpa & Neha Mahto entitled **“Farm2Door”** in partial fulfilment of the requirements for the degree of Computer Application is recommended for final evaluation.

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SIGNATURE

Mr. Janak Lal Karn

SUPERVISOR

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LETTER OF APPROVAL

This is to certify that this project prepared by Neha Mahto (TU Roll No. 93902175) and Pasang Tasi Sherpa (TU Roll No. 93902177) entitled **"Farm2Door"** in partial fulfilment of the requirements for the degree of Bachelors of Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

Signature of Supervisor ____________________ Mr. Janak Lal Karn Lecturer Department of IT Orchid International College	**Signature of HOD/ Coordinator** ____________________ Er. Dhiraj Kumar Jha Head of Department Department of IT Orchid International College
Signature of Internal Examiner ____________________ Mr. Naresh Yadav Project Coordinator Department of IT Orchid International College	**Signature of External Examiner** ____________________

ABSTRACT

Farm2Door is a web-based application that connects farmers directly to consumers through a marketplace. This approach removes middlemen from the traditional agricultural supply chain. The system seeks to promote fair pricing, boost farmer's profits and give consumers easy access to fresh agricultural products. The platform offers key features, such as user registration and login, product management for farmers, product browsing and ordering for consumers and admin controls for managing users, product, orders and generating reports. By allowing clear transactions and improving market access, Farm2Door helps change agricultural marketing for the digital age while improving efficiency and simplifying operations.

Keywords: *web-based application, fair pricing, fresh agricultural products, marketplace*

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LIST OF ABREVIATIONS

Abbreviations	**Full Form**
DFD	Data Flow Diagram
ERD	Entity Relationship Diagram
UI	User Interface
HTML	Hypertext Markup Language
CSS	Cascading Style Sheet
JS	JavaScript
PHP	Hypertext Pre-processor
SQL	Structured Query Language
MYSQL	My Structured Query Language
CASE	Computer Aided Software Engineering
Middlemen/Intermediaries	Traders or brokers who traditionally buy from farmers and resell to consumers at higher prices
Organic Produce	Vegetables grown without synthetic fertilizers/pesticides.
COD	Cash on Delivery

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Chapter 1: Introduction

1.1. Introduction

The Farm2Door platform is designed to transform the way agricultural products reach consumers by removing intermediaries and connecting farmers directly with buyers. Conventional supply chains often raise costs for consumers while lowering profits for producers. Farm2Door addresses this by offering a web-based marketplace where fresh produce can be bought and sold in a transparent and efficient manner.

Through the platform, farmers can list their products online, allowing consumers to browse and place orders from any location. The system's interface is built with HTML, CSS and JavaScript on the frontend and uses PHP with MySQL on the backend to ensure responsiveness and ease of use. Core functionalities include account registration, product management for farmers, browsing and purchasing options for consumers, administrative tools for user supervision and reporting and automated email notifications via PHPMailer.

The design prioritizes usability, ensuring both new and experienced users can navigate the platform easily. Features such as secure authentication, categorized products and order tracking simplify the purchasing experience. By promoting a direct-to-consumer model, Farm2Door ensures fair pricing that benefits farmers while providing affordable produce to consumerss. Additionally, the platform contributes to the digital advancement of agriculture and supports fair trade practices, creating a reliable and cost-effective virtual marketplace.

1.2. Problem Statement

Conventionally, the agricultural supply chain is heavily reliant on multiple layers of intermediaries. These middlemen often dominate the market, creating a significant gap between the producer and the end consumer. Due to this dependency, farmers receive reduced profit margins, while consumers are required to pay higher prices because of additional costs added throughout the supply chain. Furthermore, traditional marketplaces require physical presence for selling and purchasing agricultural products, which consumes significant time and effort for both parties. There is also a lack of a centralized digital platform where farmers can directly showcase their products and consumers can conveniently access fresh produce. These limitations highlight the need for a secure, efficient and web-based system that facilitates direct interaction between farmers and consumers while ensuring transparency and fair trade practices.

1.3. Objectives

The general objective of the project is to develop a web application that connects farmers directly with consumers for purchasing fresh produce online. The specific objectives are listed below:

- To provide a platform for farmers to list and manage their products online.
- To provide buyers with a straightforward way to access and purchase fresh agricultural produce from farmers.
- To improve the efficiency and transparency of the agricultural supply chain by eliminating intermediaries.
- To ensure fair pricing for farmers and affordable produce for consumers.

1.4. Scope and Limitations

Farm2Door allows users to register and create accounts to either sell or purchase fresh farm produce. Farmers can add, update and delete product details, while consumers can browse products categorized for easy searching and place orders directly. The system supports online payment and Cash on Delivery (COD) as the payment method and provides an admin interface for managing users and generating basic reports. The platform is designed to be simple, responsive and accessible through any modern web browser.

Some limitations are:

- No delivery tracking system.
- No recommendation system for products.
- No review or rating system for farmers or products.

1.5 Report Organization

This report is organized into distinct chapters to present the development of the Farm2Door e-commerce application in a structured manner:

- **Chapter 1:** Provides a brief introduction to the project, including objectives, problem statement, purpose and need of the system, scope and limitations.
- **Chapter 2:** Covers the background study and literature review related to e-commerce platforms and agricultural supply chains.
- **Chapter 3:** Focuses on requirement analysis, detailing both functional and non-functional requirements as specified in the Software Requirements Specification (SRS).

- **Chapter 4:** Describes the implementation and testing of the system.
- **Chapter 5:** Presents the conclusion of the project and outlines future recommendations for enhancement and deployment.

Chapter 2: Background Study and Literature Review

2.1. Background Study

Farm2Door is an e-commerce application developed to modernize the agricultural supply chain by connecting farmers directly with consumers through an online platform. In the traditional system, farmers depend on middlemen to sell their products, which often reduces their profit and increases the price for consumers. This creates problems such as unfair pricing, lack of transparency and delays in the buying and selling process. To solve these problems, Farm2Door provides a simple web-based application where farmers can register, add their vegetable products, update prices and view orders placed by consumers. Consumers can register, browse available products and place orders directly from farmers using the application. An admin user manages users, products and views basic reports like daily sales, inventory and payments report.

The application uses a database to handle information related to users, products and orders so that the system can show correct details and function properly. Only basic features are included in this project, such as product listing, order placement and simple reports.

2.2. Literature Review

Digital agriculture and online farm-to-consumer platforms have become more important in recent years. They connect farmers directly with consumers, cut out middlemen and improve market efficiency. Directly marketing farm products lets farmers secure better prices, expand their consumer base and lower post-harvest losses [1].

King et al. [1], in their study on direct marketing, showed that it helps farmers reach consumers effectively while increasing transparency in pricing and market information. Pesch [2] also pointed out that farmer-to-consumer direct marketing boosts farmer's income and gives consumers better access to fresh, high-quality produce. This approach supports local food systems and promotes sustainable consumption by allowing urban and rural consumers to buy fresh agricultural products directly from producers.

In Nepal, KC [3] studied agro-based e-commerce portals and found that traditional marketing involves several layers of intermediaries. This setup raises prices for consumers while providing farmers with little profit. Their study, which used a PHP and MySQL framework similar to Farm2Door, suggests that online platforms can help close this gap. Still, they found that although 68% of local farmers are interested in these platforms, the simplicity of the user interface is important because of differing levels of digital skills [3].

Additionally, Neupane and Thapa [4] looked into the mandarin marketing system in Nepal and questioned the traditional role of intermediaries. Their research showed that while intermediaries offer crucial logistics, they often take advantage of farmer’s weak bargaining position and lack of market knowledge. They frequently “harass and cheat” farmers with unfair pricing practices. To address this issue, they recommend establishing group marketing systems whether digital or cooperative that give farmers better market data [4].

Regarding modern delivery logistics, Lalonde and Singh [5] noted that consumer expectations and supply chain challenges, like timely delivery and product quality, are vital for the success of farm-to-consumer digital marketplaces. They argue that overcoming these challenges with secure payment systems, dependable logistics and user-friendly digital platforms is vital for broader acceptance and long-term viability of these systems.

Moreover, Gurjar et al. [6] states that modernizing the vegetable supply chain with platforms like EatFresh can tackle long-standing inefficiencies. Their study shows that providing vendors with tools for real-time inventory tracking, automated order management and route optimization cuts waste and minimizes delays. They also found that data driven analytics help stakeholders understand consumer preferences and sales trends, ensuring high service standards and sustainable business growth.

In summary, the literature suggests that digital farm-to-consumer platforms, like Farm2Door, can improve market access for farmers, cut down inefficiencies and meet consumer needs for fresh produce. At the same time, it emphasizes the importance of managing logistical and technological challenges carefully.

Chapter 3: System Analysis and Design

3.1. System Analysis

For the Farm2Door system, system analysis involves gathering and interpreting information about how farmers, consumers and administrators interact within the platform. It identifies existing problems in the traditional supply chain such as price manipulation, limited market access for farmers and lack of transparency and breaks the system into components like user management, product listing, order processing, payment handling and delivery coordination.

3.1.1. Requirement analysis

Requirement Analysis for Farm2Door was carried out to identify the needs of farmers and consumers and to translate those needs into clear, actionable requirements. Since the project aims to connect farmers directly with consumers through a web-based platform, the analysis focused on usability, fairness in pricing and system reliability.

3.1.1.1. Functional Requirements

The Farm2Door system functions according to its functional requirements which outline the essential operations required for platform operation. The system should enable farmers to manage their products and orders while consumers can use the web portal to search for products and make purchases and payments. The admin needs to manage user accounts, manage orders and generate different types of reports. The system requires secure login functions together with accurate order processing and proper stock management and dependable payment processing to enable efficient digital agricultural marketplace operations.
Some Functional Requirements of Farm2Door are:

3.1.1.1.1 Farmer Functional Requirements

- **Farmer Login:** The system shall allow a secure login page for farmers using their username and password.
- **Register Farmer Account:** The system shall allow farmers to register by providing personal information including full name, email address, location and a secure password.
- **Manage Products:** The system shall allow farmers to add new products with details (name, price, quantity, description, image), update existing products or delete them.

- **View Orders:** The system shall allow farmers to view all orders placed by consumers for their products.
- **Receive Confirmation/Error Messages:** The system shall provide clear feedback for successful actions (e.g., product added) or invalid inputs.

3.1.1.1.2. Consumer Functional Requirements

- **Consumer Login:** The system shall allow consumers to log in securely using their username and password.
- **Register Consumer Account:** The system shall allow consumers to register with personal details (name, email, address, password).
- **Browse Products:** The system shall allow consumers to view available products with details (name, image, price, stock).
- **Search/Filter Products:** The system shall allow consumers to search and filter products to find items easily.
- **Add to Cart:** The system shall allow consumers to add products to a shopping cart before placing an order.
- **Place Orders:** The system shall allow consumers to place orders, which updates the database and generates confirmation messages.
- **Receive Confirmation/Error Messages:** The system The system shall provide clear messages for successful order placement or invalid actions.

3.1.1.1.3. Admin Functional Requirements

- **Admin Login:** The system shall allow admin to log in securely with predefined credentials.
- **Manage Users:** The system shall allow admin to view, create, update and delete: farmer accounts, consumer accounts and admin accounts (predefined only)
- **Manage Products:** The system shall allow admin to delete product listings across the system.
- **View Reports:** The system shall allow admin to access automatically generated reports.

Figure 3.1: Use Case Diagram of Farm2Door

The Use Case Diagram for Farm2Door shows a number of features through which the main actors (Farmer, Consumer and Admin) use the system. At the centre of the diagram, Registration and Login processes have been set up as the main gateway to various activities depending on the actor's role. The Consumer role after being properly authenticated is granted the opportunity to engage in various marketing activities starting from searching and filtering products to finally placing an order (Place Order use case) which also includes the payment part. The Farmers keep the market updated with their products either by adding new ones or refreshing the stock. Meanwhile, Admin's functionalities encompass user management (thus ensuring the system integrity), checking and updating the product database and performing sales analysis through the consultation of the sales reports. This diagram is an excellent visual representation of the platform modular structure and user-oriented flow.

Table 3.1: Use Case for Registration Module

Field	**Description**
Use-Case ID	UC01 – Register
Primary Actor	Farmer / Consumer
Secondary Actor	System
Description (Steps)	1. User opens the registration page. 2. User enters personal details (name, email, password, role, etc.). 3. User submits the registration form. 4. System validates the entered information (e.g., checks for unique email, password strength, required fields) and stores the info.
Success Scenario	The user account is successfully created in the appropriate entity table (Farmer or Consumer).
Failure Scenario	Invalid or missing data is detected during validation, leading to registration denial. The System displays error messages next to the incorrect fields.

Table 3.2: Use Case for Login Module

Field	**Description**
Use-Case ID	UC02 – Login
Primary Actor	Authenticated User (Any Farmer, Consumer or Admin)
Secondary Actor	System

Description (Steps)	1. User provides their login credentials to access the system. 2. User submits the login form. 3. System verifies the credentials against the stored records in the respective user tables. 4. If credentials are correct, the System establishes a secure session. 5. System grants access to the appropriate dashboard based on the user's role.
Success Scenario	The user is successfully logged in, a session is started and the user is redirected to their dashboard.
Failure Scenario	Incorrect credentials are provided, leading to access denial. An error message is shown, indicating that the username or password entered does not match our records.

Table 3.3: Use Case for Add to Cart Module

Field	**Description**
Use-Case ID	UC-C04 – Add to Cart
Primary Actor	Consumer
Secondary Actor	System
Description (Steps)	1. Consumer uses Search Product or View Product to find a desired product. 2. Consumer clicks the "Add to Cart" button. 3. If available, System updates the Consumer's cart session with the product and quantity.
Success Scenario	The product is successfully added to the Consumer's shopping cart session.

Failure Scenario	Quantity Error: The requested quantity is greater than the available Stock quantity. System displays an error message: "Quantity exceeds stock."

Table 3.4: Use Case of Add Product Module

Field	**Description**
Use-Case ID	UC-F01- Add Product
Primary Actor	Farmer
Secondary Actor	System
Description (Steps)	1. Farmer accesses the system (via Login) and navigates to the "Manage Products" section. 2. Farmer clicks the **"**Add products**"** button. 3. Farmer enters all required product details (name, category, price, stock quantity, threshold etc.). 4. Farmer submits the form. 5. System validates the input (e.g., price is positive, stock is numeric). 6. System creates a new record in the Product table, associating it with the Farmer's ID.
Success Scenario	The System successfully validates the input and creates a new, active product record in the Product table.
Failure Scenario	Validation Error: The Farmer enters invalid or missing data. The System rejects the submission and displays an error message, prompting the Farmer to correct the inputs.

Table 3.5: Use Case of Manage Users Module

Field	Description
Use-Case ID	UC-A01- Manage User
Primary Actor	Admin
Secondary Actor	System
Description (Steps)	1. Admin accesses the system (via Login) and navigates to the "Manage Consumers or Manage Farmers section". 2. Admin searches or filters for a specific user (Farmer or Consumer). 3. Admin selects a user and clicks "Block" 4. Admin confirms changes. 5. System validates the update and modifies the corresponding record in the Farmer or Consumer table.
Success Scenario	The System successfully validates and applies the modifications to the target user's record in the database.
Failure Scenario	Database Error

Table 3.6: Use Case of View Reports

Field	Description
Use-Case ID	UC-A02- View Reports

Primary Actor	Admin
Secondary Actor	System
Description (steps)	1.Admin accesses the\ system (via Login) and navigates to "Reports". 2. Admin selects a report type (e.g.Daily Sales, Inventory, Payments Report) and a date range. 3. System queries the relevant database tables (Order, Payment, Product, etc.). 4. System aggregates the data and displays the metrics in a summarized report format.
Success Scenario	The System successfully processes the complex query and displays the requested financial or operational report.
Failure Scenario	Query Error: A database connection error prevents the query from running. The System displays a generic error message and asks the admin to try again later.

3.1.1.2. Non-Functional Requirements

The non-functional requirements of the Farm2Door system define the quality standards the platform must maintain during operation. The system must provide reliable performance, fast response time and secure access to protect user data and transactions. It should support multiple users simultaneously without performance degradation while ensuring system stability and minimal downtime. The platform must also offer a user-friendly interface and maintain data accuracy to ensure smooth and efficient marketplace operations for farmers, consumers and administrators.

Some Non-Functional Requirements of Farm2Door are:

3.1.1.2.1. Performance Requirements

- Web pages (home, product listings, dashboards) should load within 3–5 seconds under normal conditions.

- The system should handle multiple farmers and consumers accessing the platform at the same time without performance issues.
- Search results and product listings should be displayed quickly (within 3 seconds).

3.1.1.2.2. Security Requirements

- All user accounts (farmers, consumers, admin) must be protected with username and password authentication.
- Access control must separate farmer, consumer and admin privileges.
- Sensitive data (user information, product details, orders) must be securely stored in the database.
- Only authorized roles can manage products, view orders or access reports.

3.1.1.2.3. Usability Requirements

- The interface must be simple and easy to navigate for all users (farmers, consumers, admin).
- Pages should be mobile-friendly and responsive.
- Farmers and consumers must clearly understand their actions (e.g. product added, order placed) through confirmation messages.

3.1.1.2.4. Reliability Requirements

- The system should be available with minimum downtime during normal operations.
- The system must guarantee that no data related to users, products or orders is lost during updates or transactions.
- Regular backups of the local database should be maintained.

3.1.1.2.5. Maintainability Requirements

- The system should be easy to upgrade (PHP code, database tables, UI components).
- Modules (login, product, management order handling, reporting) should be separated to make debugging and future enhancements easier.

3.1.1.2.6. Scalability Requirements

The system should handle future expansion with:

- More farmers and consumers.
- More product categories (beyond vegetables, if required later).

3.1.1.2.7. Compatibility Requirements

- The system should run across all major browsers (Chrome, Edge, Firefox, Safari).

3.1.2. Feasibility Analysis

3.1.2.1. Technical Feasibility

The project is technically feasible because it uses mature, well-supported technologies. The system is built with HTML, CSS, JavaScript, PHP and MySQL, all of which are lightweight and available on any standard web server or XAMPP.

Hardware requirements are minimal (standard PCs/laptops with 4GB RAM and internet access) and no specialized tools are required.

Thus, the system can be successfully developed and deployed using the existing technical resources.

3.1.2.2. Operational Feasibility

Acceptance of the system from farmers, consumers and admins would be easy because it simplifies tasks and reduces errors compared to manual processes. The interfaces are simple, intuitive and require only basic computer knowledge.

Farmers can register and manage products with little training, Consumers can browse and place orders instantly and admin can manage users and products with minimal effort.

Thus, the operational feasibility of the system promotes smooth functioning in day-to-day academic activities.

3.1.2.3. Economic Feasibility

Manual errors, paperwork and hidden administrative time are reduced by this solution, making it economically feasible. The technologies used (PHP, MySQL, XAMPP) are open-source and free, requiring no licensing costs. The benefits of fair pricing for farmers and affordable products for consumers far exceed the slight costs of development and deployment. Therefore, the project is economically feasible.

3.1.2.4. Schedule Feasibility

The scheduling in this project is achievable since:

- Requirements, use cases, diagrams and SRS structure are already well-defined.
- Project modules (login, product management, order placement, reporting and user management) are relatively easy to implement in phases.

- Rapid development is supported by the chosen technologies (PHP, MySQL).
- The defined scope and phased approach ensure that the system can be developed, tested and delivered within the allotted semester timeframe.

Task Name	Duration	Start	Finish
Requirement Analysis	7 days	Sat 11/1/25	Mon 11/10/25
System Design	10 days	Tue 11/11/25	Mon 11/24/25
Database Design	7 days	Thu 11/20/25	Fri 11/28/25
Backend Dev	30 days	Sat 11/29/25	Thu 1/8/26
Frontend Dev	30 days	Mon 12/15/2:	Fri 1/23/26
Auth & Security	10 days	Tue 1/13/2	Mon 1/26/26
Payment Integration	4 days	Fri 1/23/26	Wed 1/28/26
Testing & Fixing	7 days	Thu 1/29/26	Fri 2/6/26
Documentation	7 days	Sat 2/7/26	Mon 2/16/26

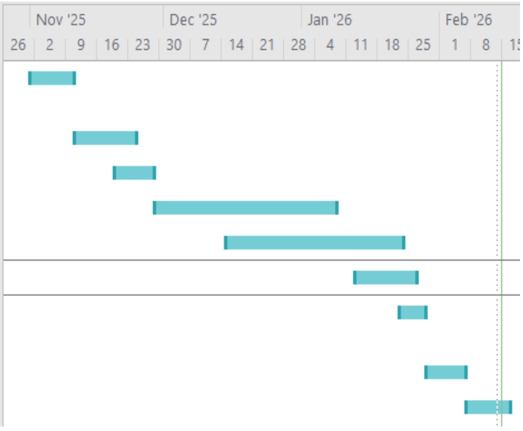


Figure 3.2: Gantt Chart of Farm2Door

This Gantt chart outlines the development timeline of the Farm2Door project, showing each phase and its duration. The project begins with Requirement Analysis to understand system needs, followed by System and Database Design to plan the structure. Backend Development and Frontend Development run next, building the core functionality and user interface. Authentication and Security ensures safe user access, while Payment Integration enables secure transactions. Testing and fixing verifies system reliability and removes errors. Finally, Documentation completes the project by recording system details and usage. The schedule demonstrates a structured workflow, overlapping tasks for efficiency and a clear progression from planning to deployment.

3.1.3. Data Modelling (ER-Diagram)

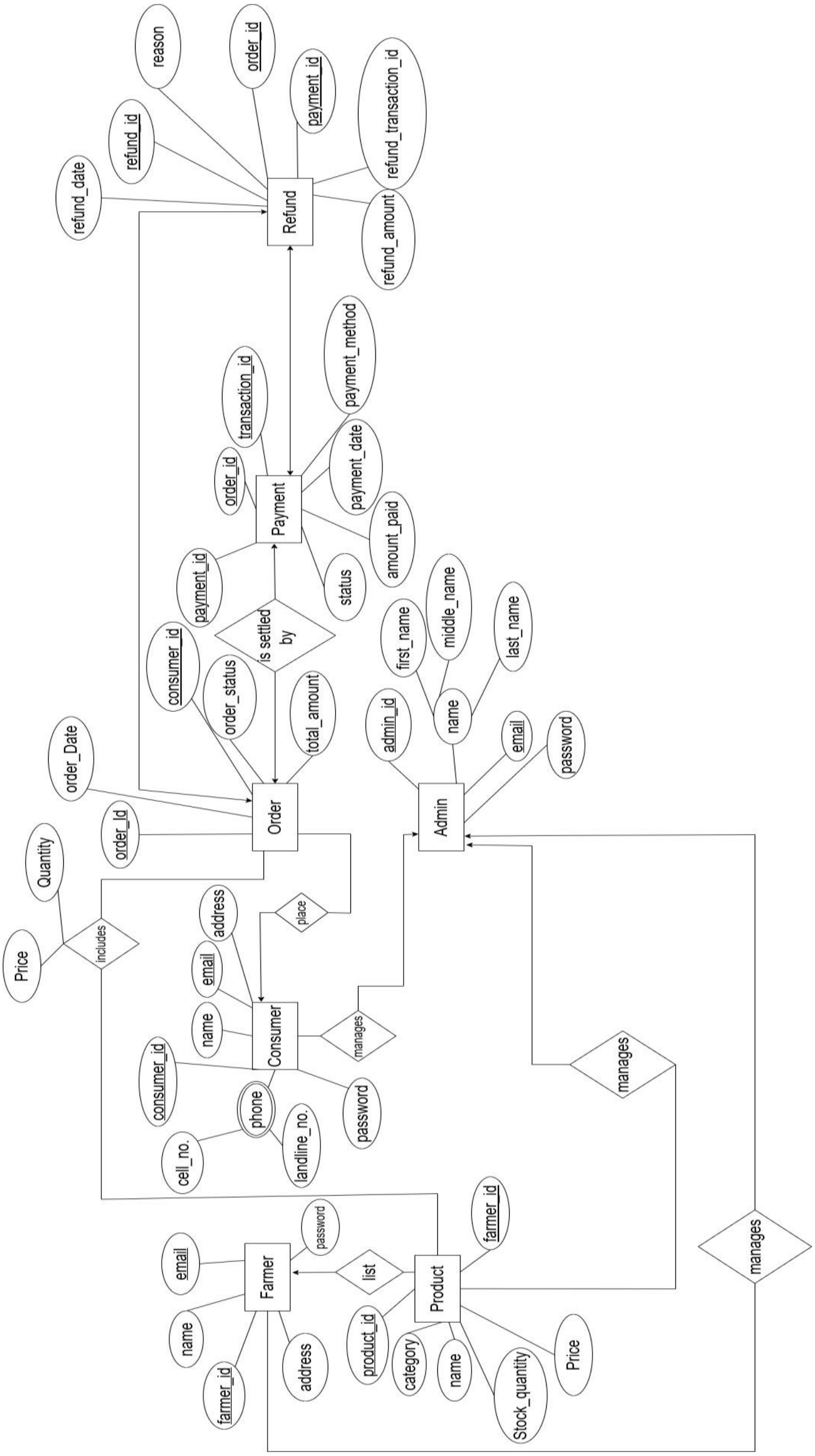


Figure 3.3: ER Diagram of Farm2Door

The Entity-Relationship (ER) Diagram of Farm2Door system basically describes the main frame of the platform's data and rules for business usage. Besides that, it lists the main entities (Farmer, Consumer, Admin and Product) with some of their attributes (IDs, names and contacts). Two main relationships are summarized by short phrases: Farmers list Products and Consumers place Orders. The ER diagram also depicts how an order can include multiple products and a payment can be made for an order to explain complex order scenarios. It further shows a Refund entity which is associated with the original orders and payments to describe the refund process. This schema not only makes it easier to logically organize all data interactions from inventory control to financial settlements, but also be able to maintain a high level of system integrity.

3.1.4. Process Modelling (Logical DFD)

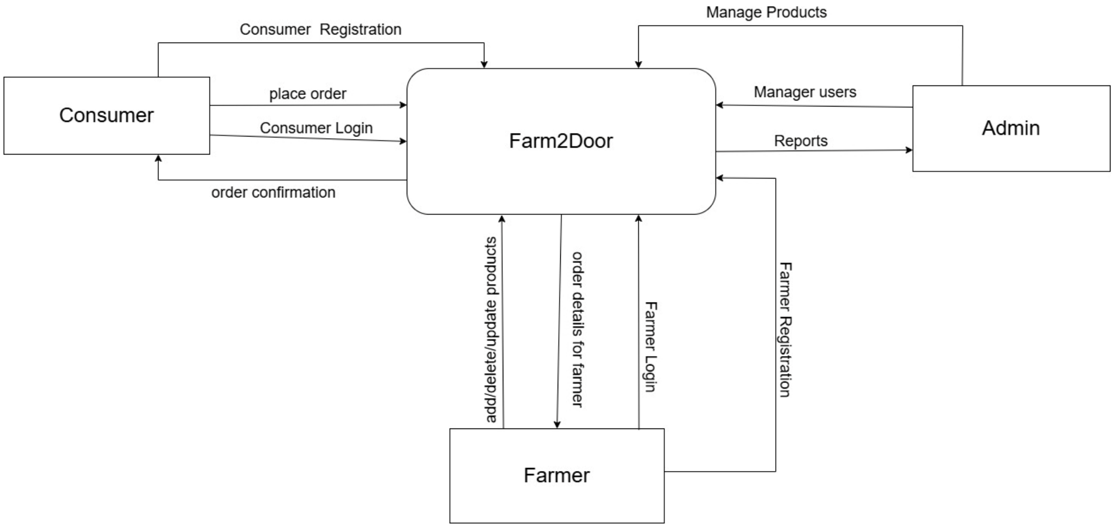


Figure 3.4: DFD Level 0 of Farm2Door

The Level 0 Data Flow Diagram (DFD), also known as the Context Diagram provides a high-level overview of the Farm2Door system's boundaries and its interactions with external entities. The diagram identifies three primary actors: Consumer, Farmer and Admin. Consumers interact with the system through registration and login processes to place orders and receive confirmations. Farmers provide product data and receive order details, while the admin manages the overall user base and generates operational reports. By centralizing these information flows, the diagram illustrates how the system acts as a digital intermediary, facilitating the seamless exchange of data between agricultural producers and end-users.

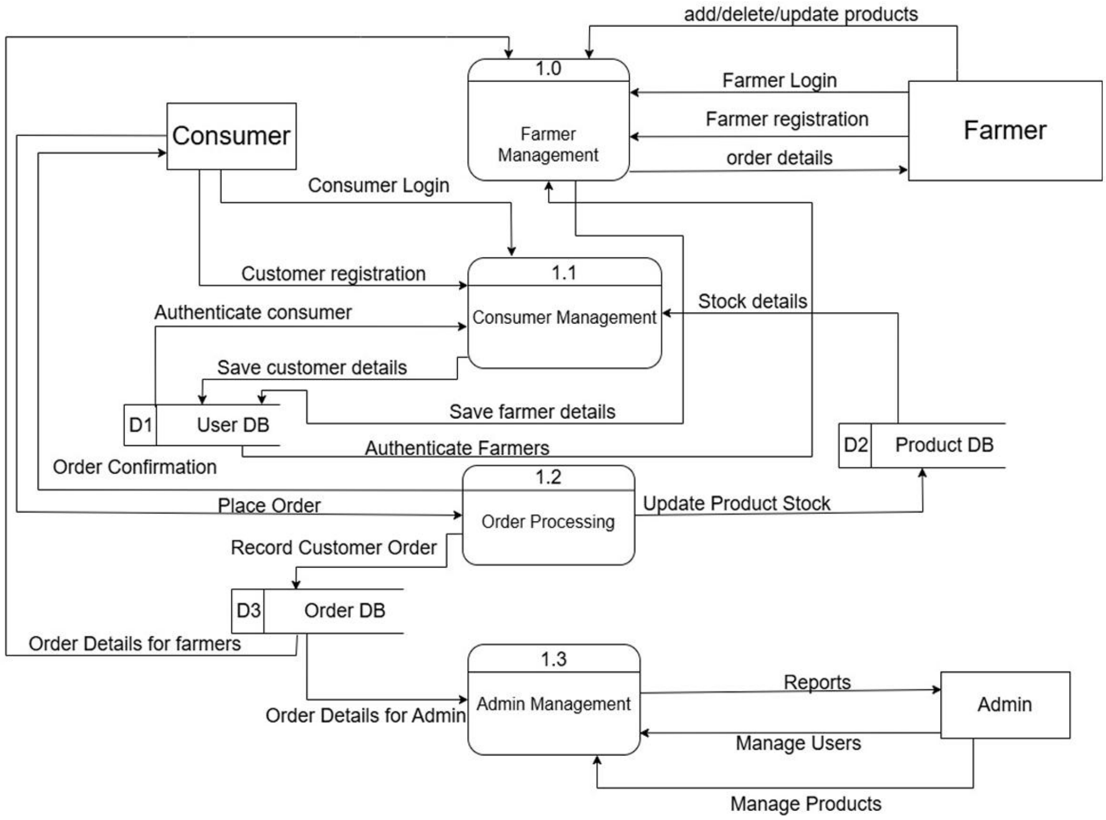


Figure 3.5: DFD Level 1 of Farm2Door

The Level 1 Data Flow Diagram (DFD) for Farm2Door decomposes the context-level process into four core functional modules: Farmer Management (1.0), Consumer Management (1.1), Order Processing (1.2) and Admin Management (1.3). This diagram illustrates how data travels between actors and specific data stores. User data is routed through management modules to the User DB (D1) for authentication and registration. The Order Processing module coordinates between consumers and farmers, simultaneously updating Product Stock (D2) and recording transactions in the Order DB (D3). Finally, the Admin Management module aggregates data from these stores to facilitate user oversight and generate operational reports, providing a detailed blueprint of the system's internal logic.

3.2. System Design

System design for Farm2Door was carried out to define the overall architecture, components, modules, data structures and interfaces required to meet the system's requirements. It transforms functional needs such as product listing, ordering, payment and delivery tracking and non-functional needs like security, performance and scalability into a clear implementation blueprint. The design organizes modules including user management, inventory control, order processing and payment integration to work together efficiently.

3.2.1. Architectural Design

Figure 3.6: Three-tier Architecture of Farm2Door

The architecture design of Farm2Door is based on three tier architecture. It allows the Farm2Door system to support separation of concerns, ease of maintainability and better scalability.

3.2.1.1. Presentation Tier

The Presentation Layer serves as the primary interface through which users interact with the Farm2Door ecosystem. This layer uses HTML, CSS and JavaScript to display all visual components which include the Farmer Dashboard and the Consumer Shopping Portal and the Admin Dashboard. The system handles user interactions because it captures user input through active elements, which it sends to the server using secure HTTPS protocols. The product listing system and order view system of the layer were built to deliver digital product information to farmers who have different levels of online skills, by making the system simple and the product information easy to understand on all devices.

3.2.1.2. Application Tier

The Application Layer operates as the primary control unit for the system which processes data through PHP code on an Apache web server. The system functions as a mediator which processes user interface requests after confirming user identity to permit authorized system operations. The system permits registered farmers to create product listings while only authorized consumers are allowed to submit purchase orders. The layer handles user verification by implementing particular rules which enable it to perform real-time stock updates, order status processing automated notification email. This layer establishes a secure area between the frontend and database which prevents public access to MySQL raw data while maintaining all system operational rules.

3.2.1.3. Data Tier

The Data Layer serves as the system base which uses MySQL Relational Database Management System (RDBMS) to achieve data consistency. The tier uses organized table structures to keep essential data which includes farmer details and product information and consumer account information and order information. The system functions as a storage bin because it enforces data integrity through relational constraints which prevent the sale of out-of-stock products and automatic order payment verification. The Data Layer manages system statistics and financial reports which function as the facility that enables Admin Dashboard users to create precise growth and profit statistics.

3.2.2. Database Schema Design

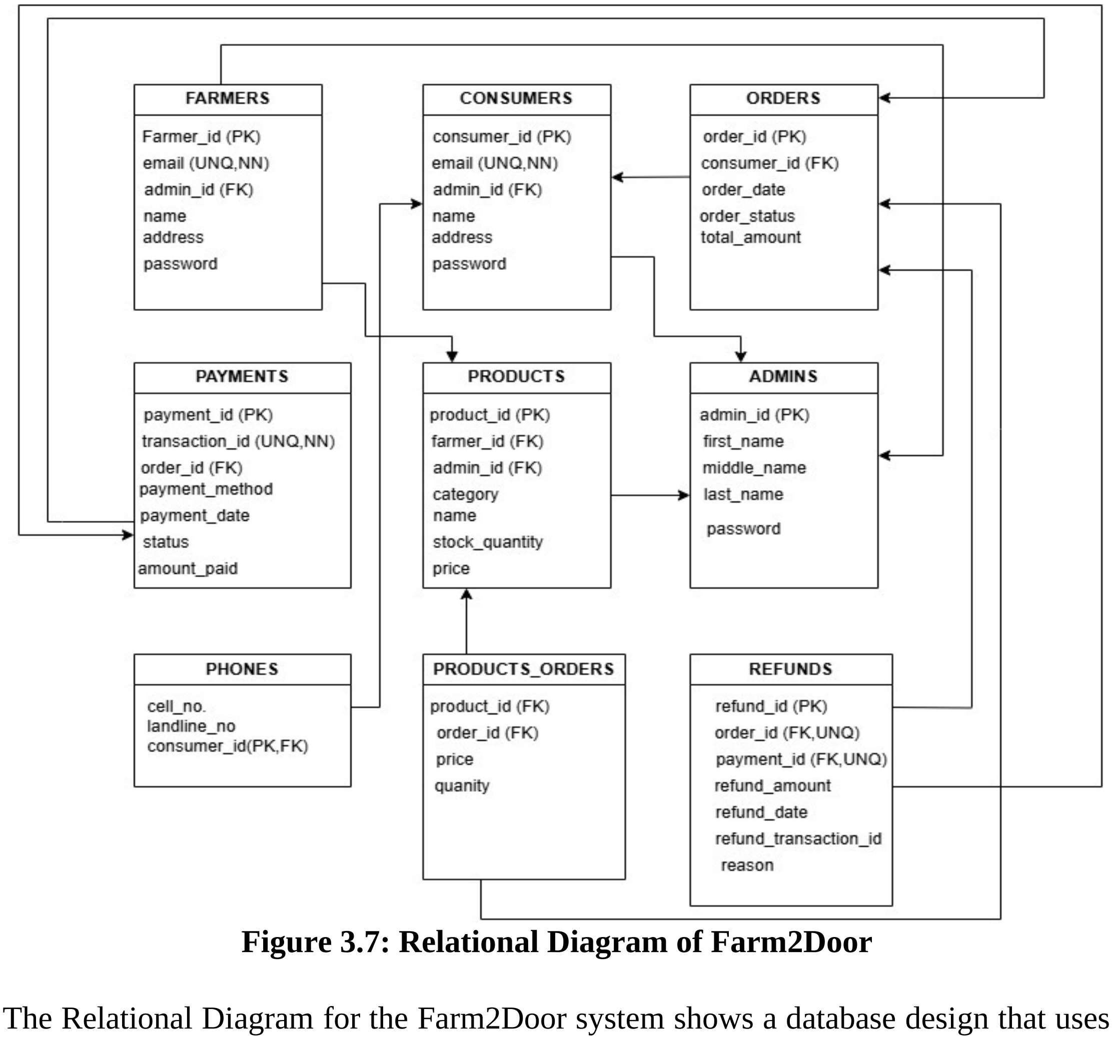


Figure 3.7: Relational Diagram of Farm2Door

The Relational Diagram for the Farm2Door system shows a database design that uses multiple tables to control access in agricultural marketplaces. The schema establishes three main user groups which include Farmers, Consumers and Admins to create a foundation for secure transaction processing. Farmers use foreign keys from the

Products table to connect with inventory management systems for accurate tracking of their inventory. The Orders and Payments tables handle sales operations by tracking both purchase times and payment. The Products_Orders table manages the relationship between items and sales while the Refunds table establishes rules for handling returned items which creates a secure system for data protection.

Table 3.7: Data Dictionary of Customer

Field Name	**Datatype**	**Constraint**
customer_id	int(11)	Primary key, AUTO_INCREMENT
firstName	varchar(100)	NOT NULL
lastName	varchar(100)	NOT NULL
Email	varchar(100)	Unique Key, NOT NULL
Password	varchar(255)	NOT NULL
Phone	varchar(15)	NULL
Address	varchar(255)	NULL
Terms	tinyint(1)	NOT NULL
created_at	datetime	DEFAULT current_timestamp()
status	enum(‘active’, ‘blocked’)	DEFAULT ‘active’
reset_token	varchar(255)	NULL
reset_expires	datetime	NULL

Table 3.8: Data Dictionary of Farmer

Field Name	**Datatype**	**Constraint**
farmer_id	int(11)	Primary key, AUTO_INCREMENT
firstName	varchar(100)	NOT NULL
lastName	varchar(100)	NOT NULL
Email	varchar(100)	Unique Key, NOT NULL
Password	varchar(255)	NOT NULL
Phone	varchar(15)	NULL
Address	varchar(255)	NULL
Terms	tinyint(1)	NOT NULL
created_at	datetime	DEFAULT current_timestamp()

status	enum('active', 'blocked')	DEFAULT 'active'
reset_token	varchar(255)	NULL
reset_expires	datetime	NULL

Table 3.9: Data Dictionary of Order

Field Name	Datatype	Constraint
order_id	int(11)	Primary key, AUTO_INCREMENT
customer_id	int(11)	Foreign Key, NOT NULL
total_amount	decimal(10,2)	NOT NULL
order_status	enum('Pending', 'Processing', 'Dispatched', 'Received', 'Ready to pickup', 'Cancelled', 'Rejected')	DEFAULT 'Pending'
order_date	timestamp	NOT NULL, DEFAULT current_timestamp()
shipping_name	varchar(100)	NULL
shipping_phone	varchar(20)	NULL
shipping_address	text	NULL
shipping_notes	text	NULL
rejection_reason	text	NULL
cancellation_reason	text	NULL

Table 3.10: Data Dictionary of Order_items

Field Name	Datatype	Constraint
item_id	int(11)	Primary key, AUTO_INCREMENT
order_id	int(11)	Foreign Key, NOT NULL
product_id	int(11)	Foreign Key, NOT NULL
farmer_id	int(11)	Foreign Key, NOT NULL
quantity	int(11)	NOT NULL
price_per_unit	decimal(10,2)	NOT NULL
subtotal	decimal(10,2)	NOT NULL

Table 3.11: Data Dictionary of Payments

Field Name	Datatype	Constraint
payment_id	int(11)	Primary key, AUTO_INCREMENT
order_id	int(11)	Foreign Key, NOT NULL
payment_method	varchar(50)	NOT NULL, DEFAULT ‘COD’
payment_status	enum(‘Pending’, ‘Paid’, ‘Refunded’)	DEFAULT ‘Pending’
transaction_id	varchar(100)	NULL
amount_paid	decimal(10,2)	NOT NULL
payment_date	timestamp	NOT NULL, DEFAULT current_timestamp()

Table 3.12: Data Dictionary of Product

Field Name	Datatype	Constraint
product_id	int(11)	Primary key, AUTO_INCREMENT
farmer_id	int(11)	Foreign Key, NOT NULL
name	varchar(100)	NOT NULL
category	varchar(50)	NOT NULL
price	decimal(10,2)	NOT NULL
stock_quantity	int(11)	NOT NULL
threshold	int(11)	NULL, DEFAULT 5
description	text	NULL
image	varchar(255)	NOT NULL
created_at	timestamp	NOT NULL, DEFAULT current_timestamp()

Table 3.13: Data Dictionary of Cart

Field Name	Datatype	Constraint
cart_id	int(11)	Primary key, AUTO_INCREMENT
customer_id	int(11)	Foreign Key, NOT NULL
product_id	int(11)	Foreign Key, NOT NULL

quantity	int(11)	NOT NULL, DEFAULT 1
created_at	timestamp	NOT NULL, DEFAULT current_timestamp()

Table 3.14: Data Dictionary of Refund

Field Name	Datatype	Constraint
refund_id	int(11)	Primary key, AUTO_INCREMENT
order_id	int(11)	Foreign Key, NOT NULL
payment_id	int(11)	Foreign Key, NOT NULL
refund_transaction_id	varchar(100)	Unique Key, NOT NULL
refund_amount	decimal(10,2)	NOT NULL
reason	varchar(255)	NOT NULL
refund_date	timestamp	NOT NULL, DEFAULT current_timestamp()

Table 3.15: Data Dictionary of Contact us

Field Name	Datatype	Constraint
id	int(11)	Primary key, AUTO_INCREMENT
Name	varchar(50)	NOT NULL
Email	varchar(50)	NOT NULL
Message	varchar(255)	NOT NULL

3.2.3. Interface Design

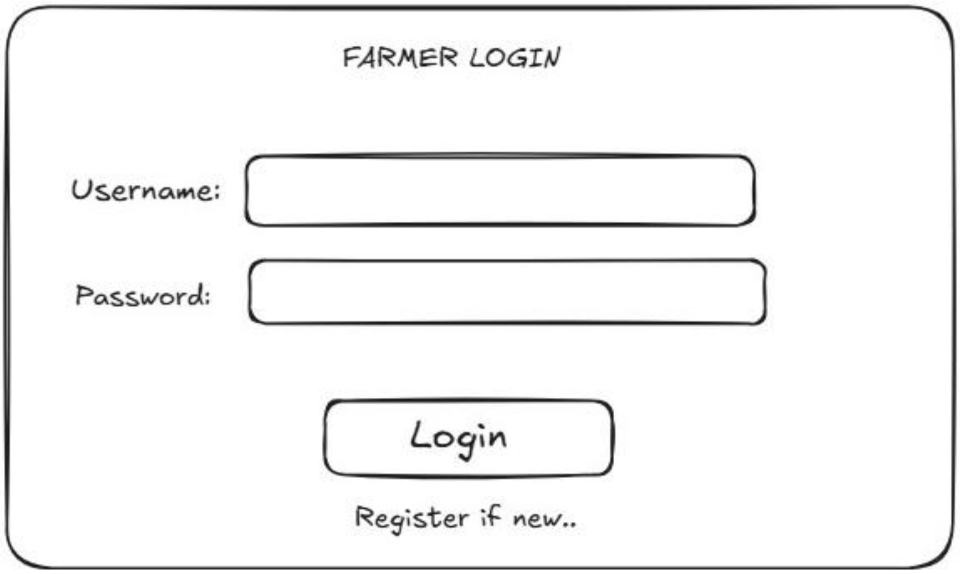


Figure 3.8: UI Design of Farmer Login

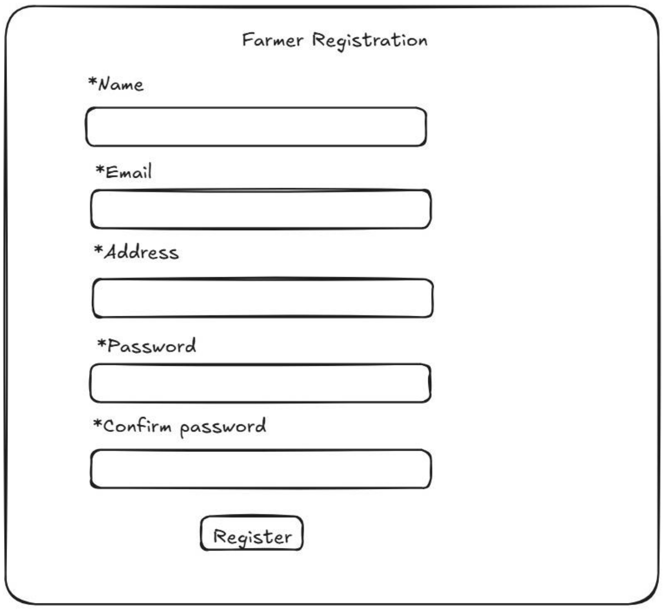


Figure 3.9: UI Design of Farmer Registration

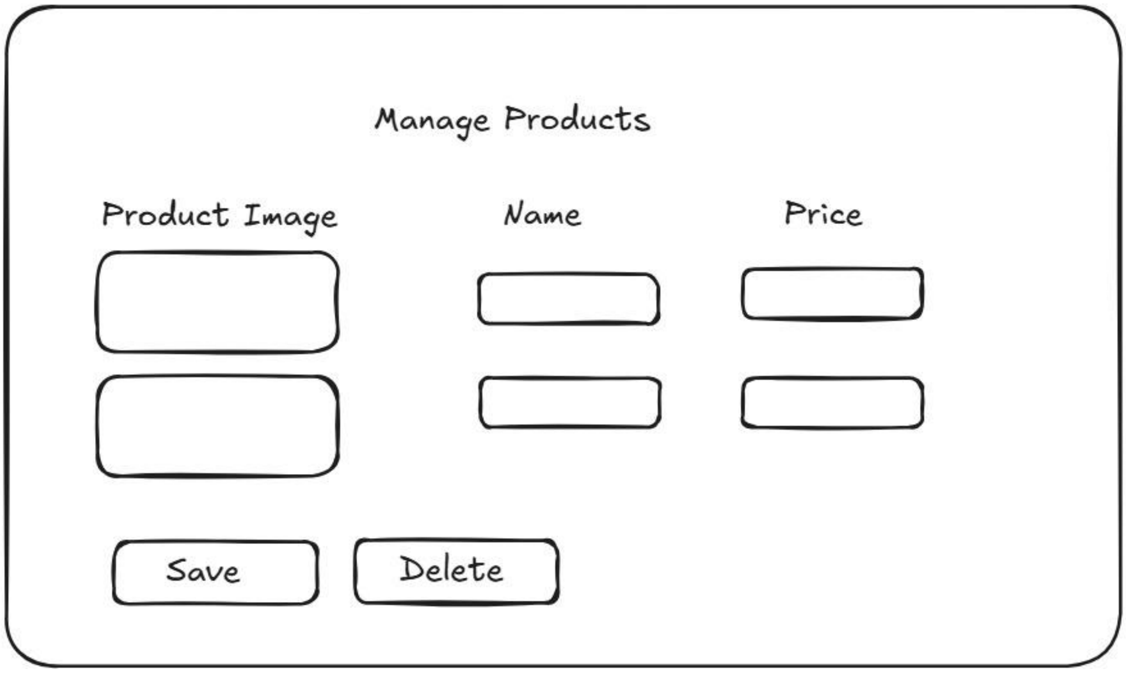


Figure 3.10: UI Design for Manage Products

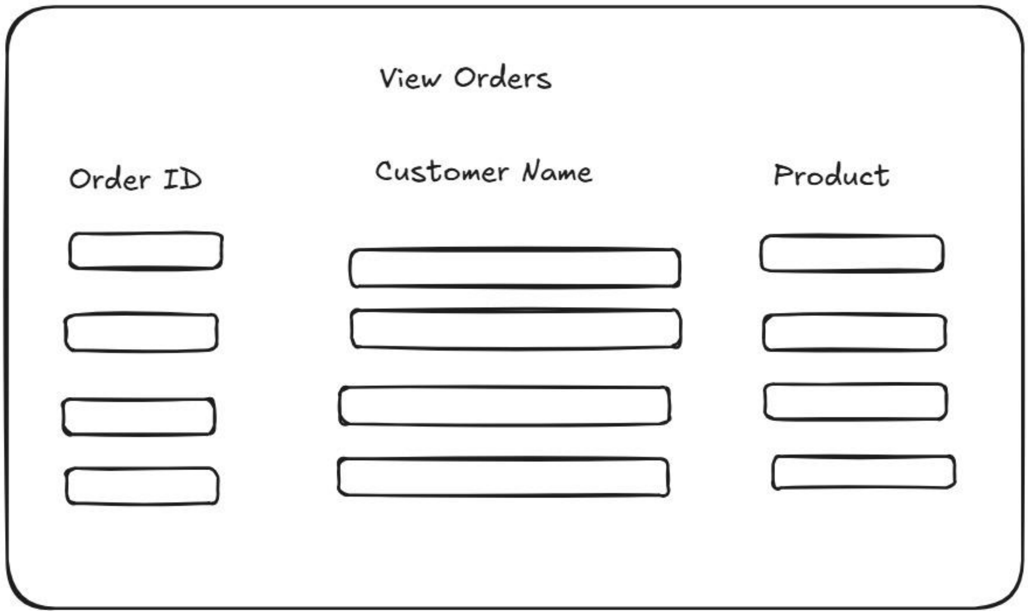


Figure 3.11: UI Design of View Orders

3.2.4. Physical DFD

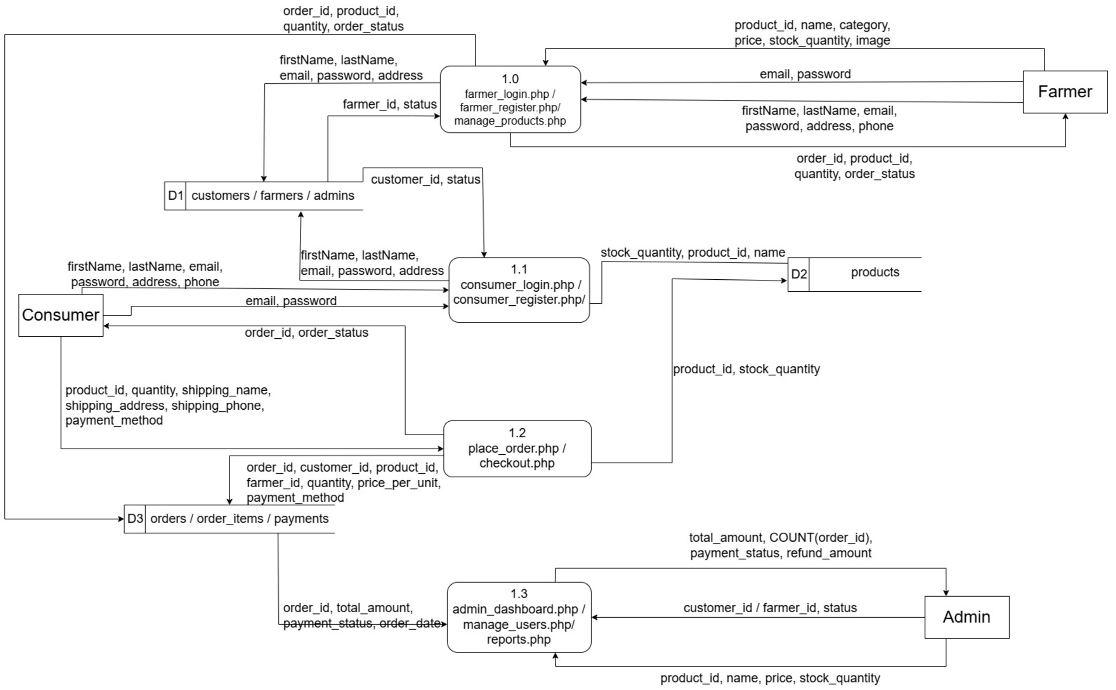


Figure 3.12: Physical DFD of Farm2DOor

The Physical Data Flow Diagram of Farm2Door illustrates the actual implementation of the system using PHP files and database tables. The diagram consists of four processes:farmer_login.php/farmer_register.php/manage_products.php(1.0),consumer _login.php/consumer_register.php (1.1), place_order.php/checkout.php (1.2) and admin_dashboard.php/manage_users.php/reports.php (1.3). Three data stores are used: D1 (consumers/farmers/admins), D2 (products) and D3 (orders/order_items/payments). Farmers submit credentials and product details such as product_id, name, category, price and stock_quantity to process 1.0. Consumers register and login through 1.1, then place orders via 1.2 passing product_id, quantity, shipping details and payment_method. Order records including order_id, customer_id, farmer_id, quantity and price_per_unit are stored in D3. The admin retrieves total_amount, COUNT(order_id), payment_status and refund_amount through reports.php for monitoring and decision-making.

Chapter 4: Implementation and Testing

4.1. Implementation

In this stage, design specifications are translated into an operational and stable software solution. This is where the system is developed. On receiving the system design documents, the work is divided in modules and actual coding is started. It is followed by testing. Several tools are used in this phase of the software development.

4.1.1. Tools Used

The following tools and technologies were used for the implementation of the system:

- CASE Tools: Figma (Used for designing user interface prototypes and wireframes, ensuring a clean and intuitive layout), Draw.io (Utilized for creating DFDs, Entity-Relationship Diagrams (ERDs) and UML diagrams to represent system design), VS Code (Used for coding), Git and Github (for tracking changes in code and backup of project).
- Programming Languages: HTML, CSS, JavaScript (Frontend), PHP (Backend), HighCharts (for visualizing data).
- Database Platform: MYSQL (Used to store and manage user data, product details and order information).

4.1.2. Implementation Details of Modules

4.1.2.1. Authentication Module

This module is intended for user login and role-based access. Credentials are validated against the user database and the system routes the user to the appropriate interface according to the roles: Farmer, Consumer or Admin.

Algorithm:

Step 1: Start

Step 2: Open login interface.

Step 3: User enters username and password.

Step 4: Validate credentials with user database.

Step 5: If credentials are valid create session and store user role.

Step 6: Redirect user to role-based dashboard (Farmer /Consumer / Admin).

Step 7: If invalid display error and request re-login.

Step 8: Terminate session on logout, browser close, or session timeout.

Step 9: End

4.1.2.2. Product Management (Farmer)

The Product Management module allows farmers to add, update and delete product details such as name, price, quantity and description.

Algorithm:

Step 1**:** Start.

Step 2**:** Authenticate Farmer and load the Farmer Dashboard.

Step 3: Input product details (Name, Category, Price, Stock Quantity).

Step 4: Execute the database command to Add, Update, or Delete records in the Product DB.

Step 5: Synchronize changes with the display list to reflect real-time inventory.

Step 7: End

4.1.2.3. Order Management (Farmer)

The Order Management module enables Farmer to manage the incoming orders from consumers. It involves accepting or rejecting orders and dispatching order to the Farm2Door Collection hub.

Algorithm

Step 1: Start

Step 2: Farmer logs into the dashboard.

Step 3: Open the Order section.

Step 4: System retrieves pending orders from the database.

Step 5: Display order details including product name, quantity, price and consumer information.

Step 6: Farmer selects an order for review.

Step 7: Farmer chooses to accept or reject the order.

Step 8: If the order is accepted, update the order status to processing.

Step 9: Send acceptance notification to the consumer.

Step 10: After farmer dispatches the order to Farm2Door Collection Hub

Step 11: Update order status to dispatched.

Step 12: If the order is rejected, update the order status to rejected.

Step 13: End

4.1.2.4. User Management (Admin)

The User Management module allows the admin to manage farmer and consumer accounts. It allows admin to block consumer and farmers which prevents them to access the login feature. All the products of blocked farmers are invisible to consumers.

Algorithm:

Step 1: Start

Step 2: Admin logs into the admin dashboard.

Step 3: Open the User Management section.

Step 4: System displays the list of registered farmers and consumers.

Step 5: Admin selects a user account.

Step 6: Admin chooses to block or unblock account.

Step 7: If blocking a user, change account status to blocked.

Step 8: Prevent blocked users from accessing the login system.

Step 9: If the blocked user is a farmer, hide all their products from consumer listings.

Step 10: End

4.1.2.5. Reports (Admin)

The Analytics and Reporting module processes order and product data for analysis and visualization. Reports such as daily Sales, product stock levels, payments report are generated automatically after putting start and end date and displayed on the admin dashboard with the help of highcharts.

Algorithm:

Step 1: Start

Step 2: Admin logs into the admin dashboard.

Step 3: Open the Analytics and Reporting section.

Step 4: Enter the start date and end date for the report.

Step 5: Submit the request for report generation.

Step 6: System retrieves order, product and payment data from the database within the selected date range.

Step 7: Process the retrieved data to calculate totals and summaries.

Step 8: Generate reports including daily sales, product stock levels and payment summaries.

Step 9: Convert processed data into charts and tabular formats for visualization.

Step 10: Display the generated reports on the admin dashboard.

Step 11: Allow admin to refresh or change date range to regenerate reports.

Step 12: End

4.2. Testing

The testing process shows that the Farm2Door platform functions properly while it meets all of its established performance standards and operational requirements. The team conducts assessment of all system modules which include Authentication and Product Management and Order Processing to evaluate their accuracy and reliability and usability.

4.2.1. Test Cases for Unit Testing

Unit testing process evaluates each software component of the Farm2Door system through tests on its individual classes and methods to confirm proper functioning in isolated tests. The testing process evaluates systems main components Authentication, Product Management, Order Processing and Report Generation through separate tests which verify their ability to process inputs and interact with the database according to designed specifications. Steps followed in performing unit tests:

- Conduct the code execution test
- Identify and resolve any errors
- Determine that the test is complete

4.2.1.1. Authentication Module

Pre-conditions: User should be Registered

Table 4.1: Test Case for Login Module

Test Case ID	Test Description	Test Steps	Input Data	Expected Result	Actual Result	Status
TC_F2D_01	Valid Consumer Login	1. Open Login Page. 2. Enter valid credentials	Username: sherpajack3@gmail.com, Password:	Should be redirected to Home page	Consumer redirected to Home page	Pass

		3. Click Login	Tashi@001		successfully	
TC_F2D_02	Invalid Consumer Login	1. Open Login Page. 2. Enter wrong credentials 3. Click Login	Username: consumer1, Password: wrongpass	Should display Error message as “Invalid Email or Password” displayed	Showed “Invalid Email or Password”	Pass
TC_F2D_03	Valid Farmer Login	1. Open Login Page. 2. Enter valid farmer credentials 3. Click Login	Username: mahtoneha2555@gmail.com, Password: farm123	Should be redirected to farmer dashboard	Farmer redirected to farmer Dashboard	Pass
TC_F2D_04	Invalid Farmer Login	1. Open Login Page. 2. Enter wrong farmer credentials 3. Click Login	Username: farmer1, Password: wrongpass	Should show error message “Invalid Email or Password” displayed	Showed “Invalid Email or Password”	Pass
TC_F2D_05	Valid Admin Login	1. Open Login Page. 2. Enter valid admin credentials. 3. Click Login	Username: mahtoneha255@gmail.com, Password: Neha@001	Should be redirected to admin dashboard	Admin redirected to admin dashboard	Pass

4.2.1.2. Product Management Module:

Pre-conditions: User Should be logged in

Table 4.2: Test Case for Product Management Module

Test Case ID	Test Description	Test Steps	Input Data	Expected Result	Actual Result	Status
TC_F2D_PM_01	Add Product with Valid Data	1.Open Add Product section. 2. Enter valid details. 3. Click add product.	Product Name: Tomato Price: 80 Quantity: 50 kg Description: Fresh Tomatoes Image: Tomatoes image	Product should be added successfully	Product added successfully	Pass
TC_F2D_PM_02	Add Product with Missing Name	1.Open Add Product section. 2.Leave product name Empty. 3. Click add product.	Product Name: Price: 80 Quantity: 50 kg Description: Fresh Tomatoes Image: Tomatoes image	Should display Product name is required	Displayed Product name is required	Pass
TC_F2D_PM_03	Add Product with Invalid Price	1.Open Add Product section.	Product Name: Tomato Price: -80	Should display valid price is required	Displayed valid price required	Pass

		2.Enter Negative Price. 3. Click add product.	Quantity: 50 kg Description: Fresh Tomatoes Image: Tomatoes image			
TC_F2D_PM_04	View Product List	1. Open View/Update Products section	Not required	Should display Product list	Product list displayed	Pass
TC_F2D_PM_05	Update Product Details	1. Open View/Update Products section 2. Select product. 3. Edit details. 4. Click Update.	Price: 90	Should display Product updated successfully	Displayed Product updated successfully	Pass
TC_F2D_PM_06	Delete Product	1. Open View/Update products section. 2. Select product. 3. Click Delete.	Not required	Should display Product deleted successfully	Displayed Product deleted successfully	Pass

4.2.1.3. Report Generation Module:

Pre conditions: Admin is logged in

Table 4.3: Test Case for Report Generation Module

Test Case ID	Test Description	Test Steps	Input Data	Expected Result	Actual Result	Status
TC_F2D _DS_01	Generate Daily Sales Report	1. Open Reports section. 2. Select Daily Sales report. 3. Input the start and end date 4. Click Generate	Start date: 2025/12/1 End date: 2026/2/1	Total daily sales within the specified date should be displayed	Displayed total daily sales within the specified date	Pass
TC_F2D _PR_02	Generate Payments Report	1. Open Reports section. 2. Select Payments Report. 3. Input the start and end date 4. Click Generate	Start date: 2025/12/1 End date: 2026/1/1	Should displayed total Paymets received, Total Refunds And net payments received	Successfully Displayed total paymets received, Total Refunds And net payments received	Pass

4.2.2. Test Case for System Testing

The system testing process assesses the full system functionality together with the complete system integration of all system components to confirm they function as a single system. The Farm2Door system undergoes complete system testing to confirm its compliance with all functional requirements which include end-to-end order processing from Farmer to Consumer and all non-functional requirements which include security measures and system performance standards.

Table 4.4: Test Case for System Testing of Farm2Door

Test Case ID	Test Description	Test Steps	Input Data	Expected Result	Actual Result	Status
TC_F2D_SYS_01	Farmer Product Management	1. Login as Farmer. 2. Navigate to 'Add Product'. 3. Fill details and submit.	Valid Farmer Credentials Item: "Fresh Tomato" Stock: 50kg Price: Rs. 60	Product should be saved in database and should appear on the products page.	Product successfully added and visible in products section	Pass
TC_F2D_SYS_02	Consumer buys product	1. Login as Consumer. 2. Add "Tomato" to cart.	Consumer Credentials: Email : sherpajack3@gmail.com Password:	Order should be placed successfully, inventory should be updated	Order placed successfully, Inventory Updated successfully, Sales report	Pass

		3. Proceed to Checkout.	Tashi@001	successfully , sales report updated successfully	updated successfully	
TC_F2D_SYS_03	Sales report update after order	1. Complete an order. 2. Login as Admin. 3. Open 'Sales Report'.	Admin Credentials: Email: mahtoneha255@gmail.com Password: Neha@001	The Admin dashboard should include the new order in the total sales and revenue calculation.	Sales report accurately included the recent order.	Pass
TC_F2D_SYS_04	Farmer Stock Update and display order details	1. Farmer logs in. 2. Views 'My Products'. 3. Checks 'Stock' column. 4. View Orders	Farmer Credentials: Email: mahtoneha2555@gmail.com Password: Neha@001	Stock should update after each sale and placed orders should appear in the received section.	Stock count was updated correctly in the farmer panel and received section in orders shows order data.	Pass

Chapter 5: Conclusion and Future Recommendations

5.1. Lesson learnt/Outcome

The journey of developing the Farm2Door application has been a significant learning experience that strengthened our understanding of building a practical digital marketplace. We discovered how crucial it is to design systems with role-based access, ensuring that farmers, consumers and administrators interact with the platform according to their responsibilities. We also realized the importance of creating intuitive user interfaces that simplify tasks such as product listing, browsing and order placement.

Another key outcome was gaining hands-on experience with system modeling tools like DFDs and ER diagrams, which helped us visualize workflows and relationships clearly.

Overall, the project not only enhanced our technical skills but also gave us insight into how technology can empower farmers, streamline consumer purchasing and support fair trade practices in agriculture.

5.2. Conclusion

This system has been effective in addressing the challenges of traditional agricultural trading by providing a faster, more accurate and organized way for farmers and consumers to interact. Products are readily available for consumers to browse and purchase, while farmers can easily manage their listings and track orders. At the same time, the system and its users can be efficiently monitored and managed by the admin. Hence, the platform not only makes processes transparent and less error-prone but also promotes fair pricing, improves accessibility and supports better decision-making in the management of agricultural trade.

5.3. Future Recommendations

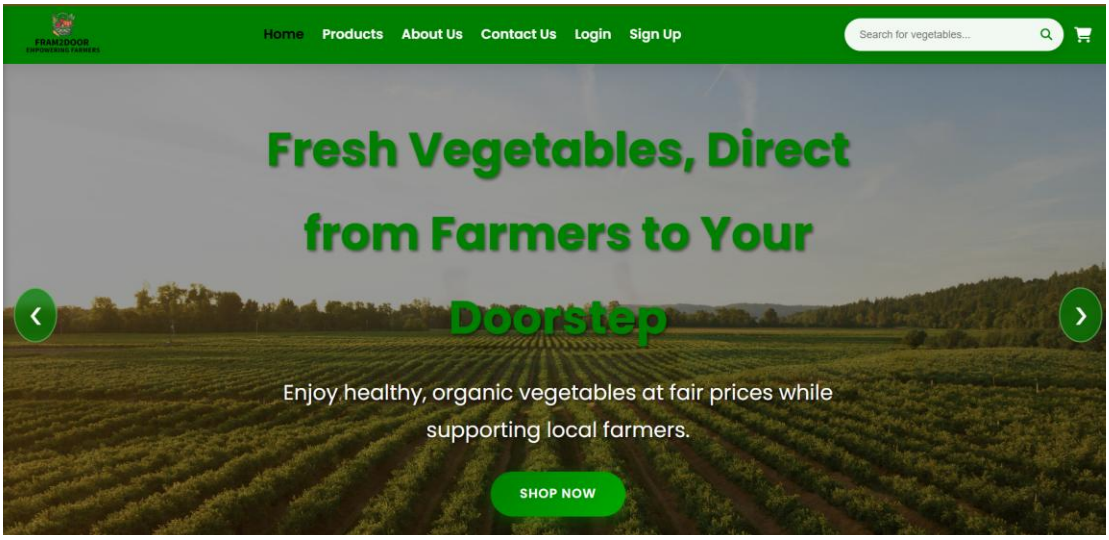
The proposed future enhancements for the Farm2Door system include:

- Mobile access applications to allow farmers and consumers to manage products and place orders conveniently from anywhere.
- Automated notifications to farmers and consumers regarding order updates, product availability and system alerts.
- Multilingual support to make the platform more accessible to local farmers and consumers in their native languages.

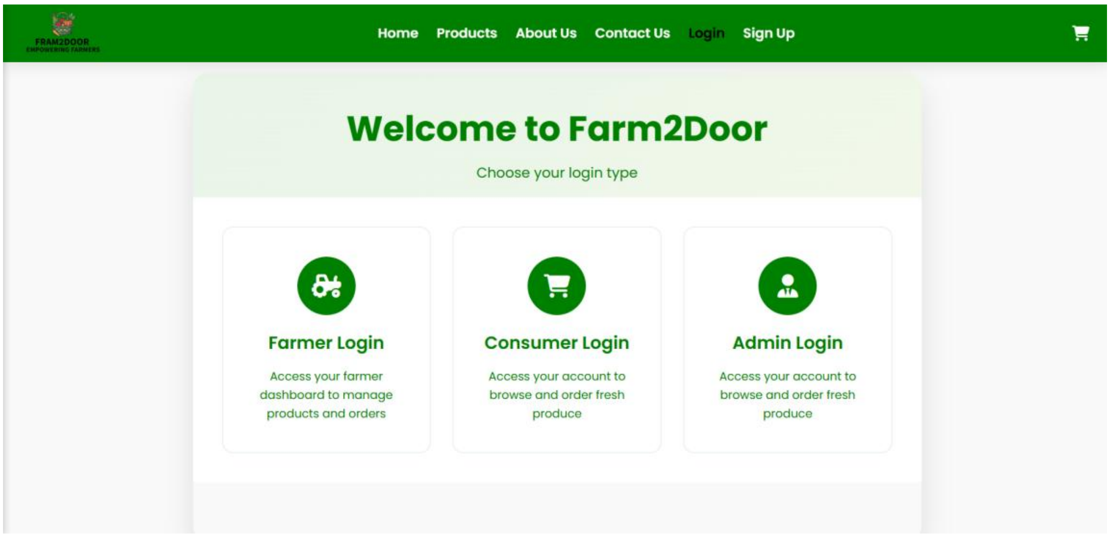
- Advanced reporting tools for admins to analyze sales trends, monitor performance and support better decision-making.

Appendices

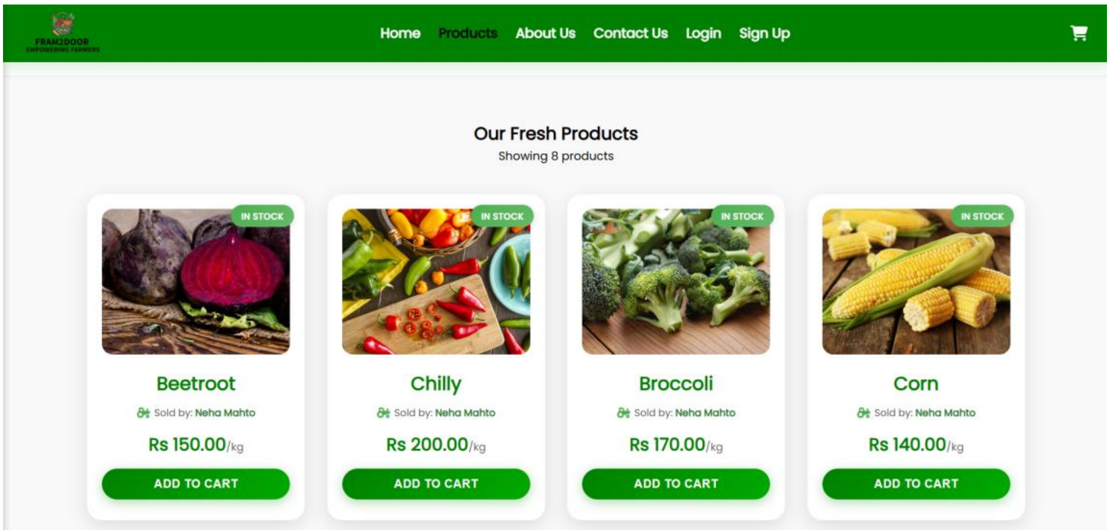
i. Home Page



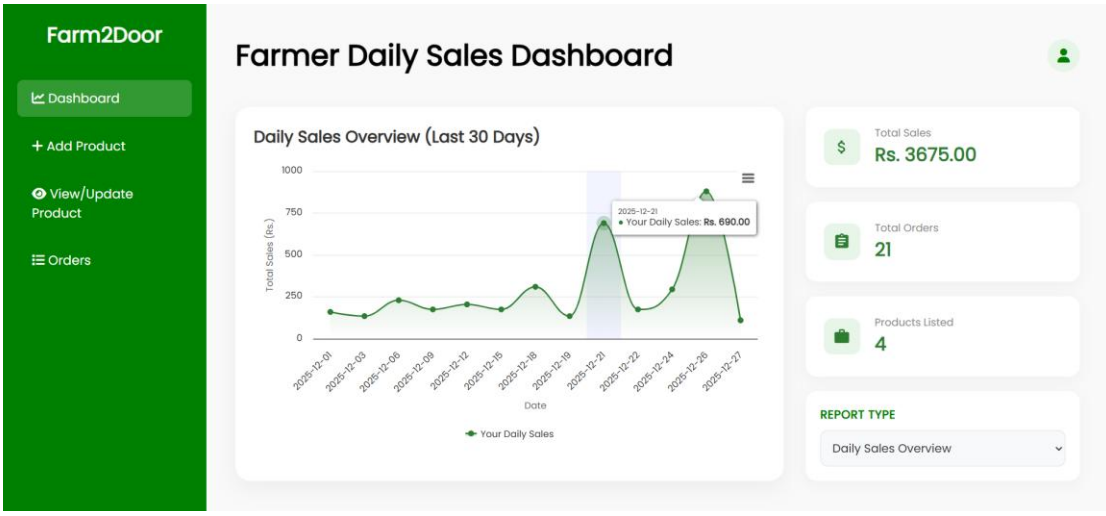
ii. Farmer Login Page



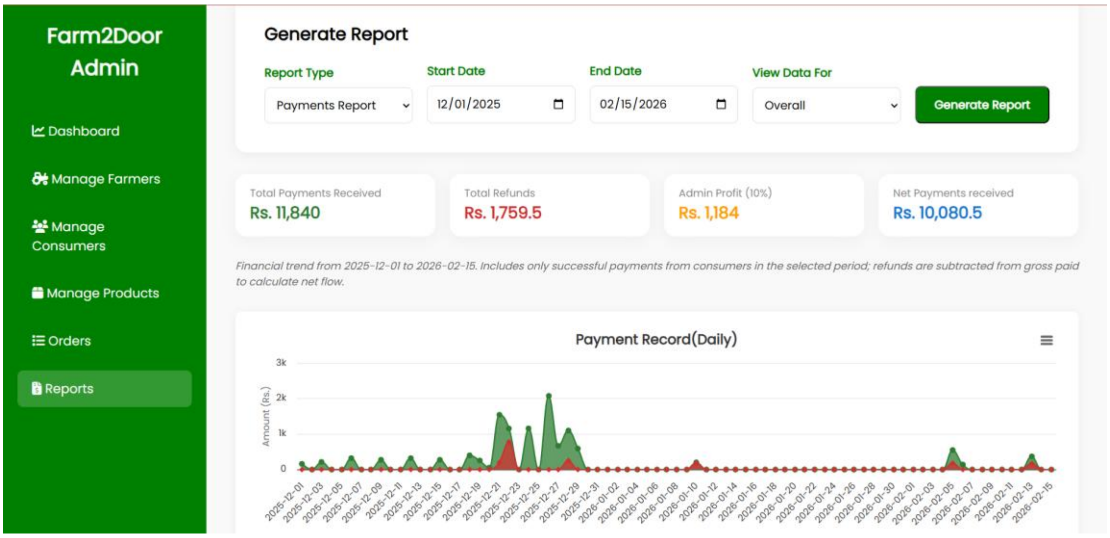
iii. Products Page



iv. Farmer Dashboard



v. Admin Reports Section



vi. Source Code

```php
<?php

$host = 'localhost';
$username = "root";
$password = "";
$dbname = "farm2door";

$conn = new mysqli($host,$username, $password, $dbname);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}

?>
```

Log Book Entry Sheet

Meeting No: #8.

Date: 25/11/2025.

Start Time:

Finish Time:

Discussion Topics:

ER diagram, Use case Diagram and UI interfaces.

Achievements:

- prototypes are well designed yet logical connectivity is missing (What comes before what- feature)

→ Prioritize — ① - - - Entry/update/delete/Report
② - - - " " " "
③ -

Problems (if any):

→ will the farmer deposit the items at our collection store or we will send the vehicle to collect the items at their farm/doorstep?

→ will we deliver the items to the customer at their door step (charges applied) or they must collect the items from our collection hub?

Tasks for Next Meeting:

→ If online payment is integrated, how can we proceed if the order is cancelled?

Student Name:

Neha Mahto

Pasang Tasi Sherpa

Supervisor Signature:

25/11/25

→ Next :— Prepare the relational model / Descriptive use-case

Log Book Entry Sheet

Meeting No: #2

Date: 12/17/2025

Start Time:

Finish Time:

Discussion Topics:

Descriptive Use Case
ER-Modified
Relational Model

Achievements:

- Discussed & documented Descriptive Use case in detail
- Drafted ER model with entities & relationships.
- Designed initial Relational Model mapping entities to tables

Problems (if any):

- Formatting issues in Descriptive Use case Tables (font & spacing)
- Attribute arrow representation in ER Model needed correction.

Tasks for Next Meeting:

- correct formatting issues in Descriptive use case tables & fix attribute arrow representation in ER diagram.
- Work pratically on project to add functionality (implementing features)

Student Name:

Pasang Tasi Sherpa
Neha Mahto

Supervisor Signature:

Log Book Entry Sheet

Meeting No: #3

Date: 19 December 2025

Start Time:

Finish Time:

Discussion Topics:

Project Functionality Addition.

Achievements:

Database Connections

Dynamic Implementation in Project

Problems (if any):

~~Hig~~ High chart JS or D3 JS

Tasks for Next Meeting:

Payment Integration. (Esewa or Stripe)

Student Name:

Pasang Tasi Sherpa

Neha Mahto

Supervisor Signature:

19/12/2025

Log Book Entry Sheet

Meeting No: #4 **Date:** 21st December 2025

Start Time: 10 : 20 **Finish Time:**

Discussion Topics:

Payment Integration & Orders

Achievements:

Payment Integration (Esewa) is achieved in such a way that when paid online, in database we see transaction id of the respective product as well as we see detail description in to pay & to receive section.

Problems (if any):

~~D3~~ (High Charts)

view Orders for Admin, farmers & consumers

Tasks for Next Meeting:

D3 High Charts.

· Make Orders page for Admin, farmers & consumers.

Student Name:

Pasang Tasi Sherpa

Neha Mahto

Supervisor Signature:

21/12/25

Log Book Entry Sheet

Meeting No: #5

Date: 22 Dec 2025

Start Time:

Finish Time:

Discussion Topics:

States of orders
Refund
Automated Email

Achievements:

Consumers Order Page was made successfully such that consumers can see what they ordered. ~~Farmers~~ Added manage products & consumers section in admin dashboard so that they can manage all products and consumers

Problems (if any):

order should be received by admin and customer will receive the product themselves from Farm2Door Central Point

Tasks for Next Meeting:

Order section for farmer & admin.

Student Name:

Pasang Tashi Sherpa
Neha Mahta

Supervisor Signature:

24/12/25

s

Log Book Entry Sheet

Meeting No: #6

Date: 2025/12/23

Start Time: 10:30

Finish Time: 12:30

Discussion Topics:

Discussed the order processing workflow and refund handling mechanism with the instructor, focusing on order status management, cancellation conditions, and refund processing logic.

Achievements:

- Orders page for farmers and Admin
- Automated emails for confirm order.
- Refund options (deducting 10% service charge)

Problems (if any):

Remove the delete option for farmers and consumers by admin, instead add block and unblock.

Tasks for Next Meeting:

Adding block and unblock option

Student Name:

Pasang Tasi Sherpa
Neha Mahto (Absent)

Supervisor Signature:

23/12/2025.

Log Book Entry Sheet

Meeting No: #7

Date: 2025/12/24

Start Time:

Finish Time:

Discussion Topics:

Discussed about high charts and reporting feature for the project.

Achievements:

Blocking and Unblocking feature is implemented from which admin can block and unblock farmers and consumers.

Reason for blocking (Business logic) must be explained.

Problems (if any):

Tasks for Next Meeting:

High Charts and Report Generation.

Student Name:

Pasang Tasi Sherpa

Neha Mahto

Supervisor Signature:

24/12/2025

Log Book Entry Sheet

Meeting No: #8

Date: 25 December 2025

Start Time:

Finish Time:

Discussion Topics:

Disc ussed handling of blocking
scenarios in system.

Achievements:

Implemented Highcharts to generate
interactive reports for farmers and
admins, including daily sales report,
stock level and top farmers report.

Problems (if any):

Future Date Disabled
Threshold
Farmer Name, status, Quantity

Tasks for Next Meeting:

Actionable Insight

Bar-group | stacked chart

Payment Report
Sales Force

Student Name:

Neha Mahto
Pasang Tasi Sherpa.

Supervisor Signature:

25/12/2025

Green Amber Red

Log Book Entry Sheet

Meeting No: #9

Date: 26 December 2025

Start Time:

Finish Time:

Discussion Topics:

Disabling Future Date, mentioning farmer's name & status

Achievements:

Impemented pagination in farmers and admin dashboard for product management section.

Problems (if any):

Actionable Insight

Tasks for Next Meeting:

Payments reporting and refunds

Student Name:

Neha Mahto

Pasang Tasi Sherpa.

Supervisor Signature:

26.12.25

Log Book Entry Sheet

Meeting No: #10

Date: 28 December 2025

Start Time:

Finish Time:

Discussion Topics:

Reviewed payment and reporting features & discussed the approach to block users rather than delete them

Achievements:

successfully implemented payments reporting, ~~and refunds~~ including total transactions and refunds and added actionable insights for stock level.

Problems (if any):

consumer and farmer should be blocked not removed

Tasks for Next Meeting:

Implement block consumer and farme feature.

Student Name:

Pasang Tosi sherpa

Neha Mahto.

Supervisor Signature:

28.12.25

Log Book Entry Sheet

Meeting No: #11

Date: 29th December

Start Time:

Finish Time:

Discussion Topics:
stripe Payment and Forget Password

Achievements:
Implemented stripe payment method along side esewa.

Problems (if any):

Tasks for Next Meeting:
Implement block consumer and farme feature and forget password mechan sm

Student Name:

Neha Mahto
Pasang Tasi Sherpa

Supervisor Signature:

Naresh
29.12.25

Log Book Entry Sheet

Meeting No: #12

Date: 5th feburary

Start Time:

Finish Time:

Discussion Topics:

Achievements:
sucessfully implemented bbck-consumer and farmer functionality

Problems (if any):

Tasks for Next Meeting:
Implement forget password mechanism

Student Name:

Neha Mahto
Pasang Tasi sherpa

Supervisor Signature:

05.02.26

Log Book Entry Sheet

Meeting No: #13

Date: 8th Febuary

Start Time:

Finish Time:

Discussion Topics:

Achievements:

Sucessfully implemented secured token based forget password funchonality via email

Problems (if any):

Tasks for Next Meeting:

Student Name:

Neha Mahto
Pasang Tasi Sherpa

Supervisor Signature:

Hamm
08.02.26

Log Book Entry Sheet

Meeting No: #14

Date: 12th Feburary

Start Time:

Finish Time:

Discussion Topics:

Achievements:
Project Completed

Problems (if any):

Tasks for Next Meeting:

Student Name:

Neha Mahto
Pasang Tasi Sherpa.

Supervisor Signature:

12.02.26

References

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